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Assignment Set – 1

Q1(a). Define unstructured data and explain its importance in business applications with suitable examples. [5 Marks]

Unstructured data is information that does not fit into a format.

This includes things like text documents, images, audio recordings, video files, social media posts and sensor data.80 Percent of all data generated globally is unstructured.

This means that companies that only use databases are missing out on most of the available information.The value of data can be seen in real-life examples.Customer reviews on websites like Amazon or Flipkart are written in language and do not follow a set format.

By analysing them with tools that understand natural language businesses can find common complaints, requests and changes in customer opinions.This information is not visible in organised sales data.For instance a company that sells electronics might find out that a certain charging cable often breaks within three months.

This information is hidden in individual reviews rather than being easily accessible in a database.The healthcare industry provides another example.Doctors notes are written in a style and medical images are made up of pixels.These do not fit into rows and columns.

Hospitals that use natural language processing on patient discharge summaries have found signs that a patient might need to be readmitted.These signs were not picked up by coded diagnosis fields.Financial services companies use tools that analyse the tone of earnings call transcripts and news articles to predict market trends.

The main idea behind these examples is that unstructured data contains information and the tools to extract this information have become practical and effective.Unstructured data has a lot of potential and organisations can use it to make decisions.

The tools to analyse data, such, as NLP, computer vision and speech recognition have improved a lot.

Q1(b). Describe the process of text preprocessing and explain the role of tokenization and lemmatization in text analysis. [5 Marks]

The text we get from the web, customer databases or document archives is full of things that get in the way. We have capital letters, punctuation, HTML things, URLs, stopwords and inconsistent spaces between words, all of which interfere with the analysis we do later on. Text preprocessing is the process of cleaning and normalizing the text so that it is in a form that algorithms can work with.

A typical process starts with making all the letters lowercase so that "Apple" and "apple" are treated as the word. We remove punctuation and special characters unless they have a meaning, like hashtags when we are analyzing social media. We also filter out stopwords which're common words like "the" "is" and "of" that do not add much value to our analysis. If the text comes from a web page we need to remove HTML tags. Javascript code as well.

The next step is called tokenization, which's where we split a long string of text into individual words. For example the sentence "Data science drives decisions" becomes a list of words: "Data" "science" "drives" "decisions". Some tokenizers, like Byte Pair Encoding, split words into smaller pieces so that we can handle a wide range of vocabulary. If we do not do tokenization we cannot count the frequency of words calculate TF-IDF vectors or do embedding calculations. It is a step for almost everything that follows.

Then there is lemmatization, which reduces words to their base form. For example "running" becomes "run" "becomes "good" and "geese" becomes "goose". This is different, from stemming which just cuts off the ends of words sometimes resulting in words that're not real. Lemmatization uses rules of language and part-of-speech tags to make sure we get a word. The result is an cleaner vocabulary, where different forms of the same word are all treated as the same word, which makes it easier to classify cluster and extract topics from the text.

Q2(a). What is a word cloud, and how does it help in understanding large text datasets? [5 Marks]

A word cloud is a way to show words from a big collection of text, where the size of each word depends on how often it appears. The more a word is used the bigger it gets. To make a word cloud you clean the text by removing words making everything lowercase and changing words to their simplest form. Then you count how often each word is used and use a tool to make the cloud. Like a Python package or an online generator.

Word clouds are really good for getting an idea of whats in a big bunch of text. For example imagine a product team gets 30,000 support tickets after an update. They can't read them all in an amount of time.. If they make a word cloud from the ticket text they might see words like "crash" "reboot" and "battery”, as the most common ones. This can point them to the problems right away.

Word clouds can also help you check if your data is clean. If you're looking at a bunch of papers and the biggest words are "http" or "doi" that probably means you missed a step when cleaning the data. Catching this early can save a lot of work on models that use data.

Word clouds do have some limitations. They only show how often words are used not how they relate to each other. They can't show phrases, context or feelings. For example "not satisfied". Satisfied" look the same in a word cloud because "not" is removed as a common word. So word clouds are a starting point but you need to do more analysis to draw conclusions.

Q2(b). Explain the TF-IDF technique and how it improves the identification of important words compared to simple frequency methods. [5 Marks]

Term Frequency – Inverse Document Frequency or Term Frequency – Inverse Document Frequency is a way to give a weight to each word in each document. It does this by combining two measures. Term Frequency – Inverse Document Frequency looks at how a word appears in a document. It also looks at how the word appears in all the documents.

Term Frequency – Inverse Document Frequency is made up of two parts. The first part, Term Frequency looks at how a word appears in one document. The second part, Inverse Document Frequency looks at how rare a word's in all the documents. Inverse Document Frequency is calculated by dividing the number of documents by the number of documents that have the word and then taking the log of that number.

When you multiply Term Frequency and Inverse Document Frequency together you get a score that rewards words that appear a lot in one document but not in others. For example a word like "the" appears in every document so its Inverse Document Frequency score is very low. This means that its Term Frequency – Inverse Document Frequency score will also be low no matter how often it appears in a document.

On the hand a technical term that appears many times in one document but not in many others will have a high Term Frequency – Inverse Document Frequency score. This is because its Term Frequency score is high and its Inverse Document Frequency score is also high. This makes Term Frequency – Inverse Document Frequency a tool for classifying documents searching for documents and extracting keywords.

Term Frequency – Inverse Document Frequency is better than counting how often a word appears because it helps to find the words that are most distinctive. It does this by giving weight to words that appear often in one document but not in many others. This makes it an useful tool, for people who work with documents and need to find the most important words. Term Frequency – Inverse Document Frequency can be used with a tool called Scikit-learn’s TfidfVectorizer, which makes it easy to use Term Frequency – Inverse Document Frequency in a few lines of code.

Q3(a). Explain the concept of text classification and its applications in real-world scenarios. [5 Marks]

Text classification is a way to figure out what a piece of text is about. It looks at the content of the text. Gives it a label. These labels are already decided, like whether something's spam or not, if someone likes something or not or what topic it is about, such as sports or politics.

There are ways of doing this. One way is to make a list of words called a TF-IDF vector and then use a computer program like Naïve Bayes or logistic regression to look at the list and decide on a label. These old ways are pretty good when there is not a lot of data to work with.

Now there are new ways that are even better. These new ways use something called learning like LSTM networks and transformer-based models such as BERT. These new ways are really good at understanding the context of the text and the order of the words. This is important for things like figuring out if someone is being sarcastic or what they want when they talk to a computer.

Text classification is used in a lot of places. For example email providers use it to filter out spam emails. Online stores use it to figure out how people feel about their products. Banks use it to look for transactions. News companies use it to decide what topic an article is about.. Companies use it to route customer complaints to the right person.

The good news is that it is getting easier to use text classification. There are -trained models, like BERT that can be used with just a few hundred examples and they can give really good results. This is a change from just a few years ago when it would take thousands of examples and a lot of work to get the same results. Text classification and models, like BERT are making it easier for companies to use this technology.

Q3(b). Describe topic modelling and explain how it helps in discovering hidden patterns in text data. [5 Marks]

Topic modelling is a way to find themes in a bunch of documents. It does this on its own without needing someone to tell it what to look for. The algorithm looks at which words appear together and finds patterns.

This is different from classification, where you need to know what you are looking for ahead of time. Topic modelling is like a discovery tool.

Latent Dirichlet Allocation or LDA is a way to do topic modelling. It was thought up by Blei, Ng and Jordan in 2003. LDA assumes that each document is made up of topics and each topic is made up of many words.

When you use LDA you tell it how topics you want to find, like ten. Then it looks at all the documents. Figures out which words are most important for each topic. It keeps doing this until it has a sense of what each topic is about.

For example if a topic has a lot of words like "patient" and "treatment" and "clinical" it is probably about healthcare. If a topic has a lot of words like "revenue" and "margin" and "growth" it is probably about business.

The person using topic modelling can then give a name to each topic, based on the words that're most important.

This is really useful when you have a lot of documents to look at. Like a law firm with 200,000 files. They can use topic modelling to group the files by what they're, about which would take a long time to do by hand.

Researchers can use topic modelling to see how ideas and topics change over time. They can look at decades of research papers. See what topics are becoming more or less popular.

Topic modelling is also used by companies that track the news to see what stories are emerging. It helps them make sense of a lot of information.

So topic modelling is a tool that helps people understand collections of documents. It is especially useful when there is much text to read.

Assignment Set – 2

Q4(a). What are the key features of MongoDB, and why is it suitable for handling unstructured data? [5 Marks]

MongoDB is a database that stores information in a way. It uses something called documents to store records. These documents can be very different from one another. This is a difference between MongoDB and other databases because it means that two pieces of information in the same group can have completely different details. For example one customer record might have a phone number. Another one might not. The database does not require that every record have the information.

MongoDB is also very good at handling a lot of information. It can spread the information across servers, which makes it easy to add more servers as the amount of information grows. This is called scalability and it means that you can just add more regular servers instead of having to buy a very expensive one. MongoDB also makes copies of the information across servers so if one server fails the information is still safe.

The database has a tool that lets you change and combine the information in many different ways. You can filter it group it. Sort it, all without having to take it out of the database and put it into a programming language like Python. MongoDB can also store files like pictures and videos right alongside the other information. This makes it very useful for teams that are working with a lot of information. MongoDB makes it easy to search and combine the information without having to set up a plan for how the information is organized. This is very helpful for teams that are working with MongoDB and need to handle a lot of information. MongoDB is a choice, for these teams because it is flexible and powerful and it can handle a lot of different types of information.

Q4(b). Explain how audio data is pre-processed and transformed into features for classification tasks. [5 Marks]

A raw audio file is a bunch of samples that were recorded at a certain speed like 16 kHz or 44.1 kHz. If you try to use this raw audio file with a classifier it does not work well because there is much information and a lot of extra noise that is not important for what you are trying to do.

First you need to get rid of some of the noise in the file using things like spectral gating or Wiener filtering so you can get rid of the background hum and static. Then you need to make sure all the audio files are at the volume so the classifier is not confused by files that are really loud or really quiet.

Next you break the file into small pieces like 20-40 milliseconds each. They overlap a little bit. You use something called a Hamming window function to make the edges of each piece smooth.

Then you take each piece. Turn it into a small set of numbers that the computer can understand. The common way to do this is by using something called Mel-Frequency Cepstral Coefficients or audio file MFCCs for short. You use a Fast Fourier Transform on each piece. Then you change the results to match how humans hear things and finally you use a Discrete Cosine Transform to get a small set of numbers. Usually 13 numbers for each piece. You can also use things like the spectral centroid, zero-crossing rate and chroma vectors to add more information to the audio file MFCCs.

The set of numbers for each piece of the audio file, where each row's a small piece of time and each column is a feature of the audio file is what you use as input for things like classifiers. The audio file classifiers can be used for things like figuring out what kind of music something is, who is speaking what kind of sounds are, in the environment and even how someone's feeling based on their voice and the audio file.

Q5(a). Define image processing and explain its significance in analyzing unstructured visual data. [5 Marks]

Image processing is about working with images. It involves changing and improving these images by looking at the pixels that make them up. A digital image is like a grid with lots of squares and each square has a value that shows how bright it is. If the image is in color it has more information with separate parts for red, green and blue.

Basic things you can do with images include making them bigger or smaller cutting out parts you do not want turning them around making them black and white adjusting how bright they are and making all the colors look even. These are usually the steps when you want to look at images because computers like to work with images that are all the same size.

There are also complicated things you can do with images like finding the edges of objects making objects look bigger or smaller separating objects from each other and looking at images in a way that shows you the patterns they contain.

Image processing is really important when you are working with images because they do not come with any information about what they show. For example a picture of a chest X-ray is a bunch of pixels until image processing helps find the important parts that a computer can use to figure out if someone has pneumonia. A picture taken from a satellite is a bunch of colors until image processing separates the different parts, like forests, water and cities into shapes that a computer can understand. Companies that make things use image processing to look at pictures from cameras on their production lines and find problems, with their products. In each of these cases image processing takes the picture and turns it into information that people can use.

Q5(b). Describe the process of image classification and its applications in real-world scenarios. [5 Marks]

Image classification is a process where we give a name to an image from a group of categories we already have. We start by getting images ready: we collect images give them names and divide them into groups to train to check and to test. We make all the images the same size, usually 224 by 224. We make sure the colors are all in the same range from 0 to 1. We also do some things to the images to make them look a little different like flipping them turning them cutting them and changing the colors so we have images to work with and the computer does not get too used to the same images.

The computer uses a kind of brain called a convolutional neural network to look at the images and give them names. The brain looks at the images in steps: it looks at the edges and the textures first. Then it looks at how all those things fit together to make objects. The brain then looks at all the things it found and makes a list of how likely it's that the image is one thing or another. The computer learns by looking at the images and trying to guess what they are. It gets better at guessing by looking at how wrong it was and trying to do better next time.

We use image classification in a lot of things. Cars can look at traffic signs and people and lines on the road. Know what they are. Doctors can look at pictures of skin. Know if something is wrong. Farmers can look at pictures of their crops. Know if they are healthy. We can even take a picture of something we like and find things that look like it.. When we put pictures on the internet the computer can look at them and know who is in them. A lot of people can use image classification now because we have computers that are already trained to do it so we do not have to start from the beginning. Image classification is used in places, like autonomous vehicles that classify traffic signs, pedestrians and lane markings in real time and dermatology apps that classify skin lesions as benign or potentially malignant and agricultural drones that classify crop regions by health status and retail platforms that enable visual search and social media platforms that auto-tag faces, in uploaded photos and all these things use image classification to make our lives easier.

Q6(a). Explain the process of video data analysis and how patterns are extracted from video sequences. [5 Marks]

Video is a sequence of images that are played one after the other. These images are called frames. They are usually captured at a rate of 24 to 30 frames per second. When we look at video it is different from looking at a picture because the video shows us how things move and change over time.

The first step in working with video is to take out the frames and select some of them to work with. We do not need to use every frame we can just use a few of them per second. Then we make these frames ready for use by resizing them making them all the same and removing any noise.

We use tools called CNNs to look at each frame and find out what is in it. This is the way we would look at a single picture.. Video also shows us how things change from one frame to the next. One way to do this is to look at how the pixels move from one frame to the next this is called flow. It tells us where things are moving and how fast they are moving.

We can also use 3D CNNs that look at many frames at the same time and find out how things are changing over time. For things that happen over a period of time like a person falling down we can use video frames and feed them into special LSTM networks that can learn what happens over several seconds.

Video is used in areas such as surveillance to detect strange behaviour sports to track how players are moving self driving cars to understand what is happening on the road and content moderation to check what is in videos. The biggest problem with video is that it makes a lot of data much more, than a picture so storing and processing this data can be very difficult.

Q6(b). Describe how machine learning techniques are used for fake news detection using textual data. [5 Marks]

Detecting news is basically a simple problem: you read an article and decide if it is real or made up. Researchers put together groups of articles that they know are true or false like the LIAR and FakeNewsNet datasets to help train computers to do this job.

They start by cleaning up the text: they make everything lowercase remove punctuation break the text into words remove common words like "the" and "and" and simplify the words to their basic form. This is where the experts get to add their ideas. They do not just look at the words in the article. Also at the style of the writing. They notice that fake news often uses emotional language has more exclamation marks, shorter sentences and uses superlatives like "best" or "worst" more often. They also notice that fake news often does not say where the information came from.

The computer can measure these things. Use them to help decide if an article is real or fake. Old computer models, like regression and random forest can get it right about 80 to 85 percent of the time. Newer models, like LSTM networks and BERT can do better because they can understand the order of the words and the context of the article.

The best systems use not the text of the article but also other information, like who posted it how fast it is spreading on social media and how people are reacting to it. This approach works well. Companies, like Facebook and YouTube use these systems to try to stop news but it is an ongoing battle because fake news is always changing.